

In [11]:

```
import numpy as np

a = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12])
b = a.reshape(2, 6, 1)
print(b.ndim)
print(b)
b=b.reshape(12)
print(b)
```

```
3
[[[ 1]
   [ 2]
   [ 3]
   [ 4]
   [ 5]
   [ 6]]

  [[ 7]
   [ 8]
   [ 9]
  [10]
  [11]
  [12]]]
[ 1  2  3  4  5  6  7  8  9 10 11 12]
```

In [37]:

```
list1=[1,2]
list2=[2,3]
list3=[4,5]
list4=[5,8]
list5=[6,9]
list6=[8,10]
list7=[2,3,4,5,6]
a=np.vstack((list1,list2,list3,list4,list5))
print(a)
a=np.delete(a,2,axis=0)
print(a)
a=np.vstack((a,list6))
print(a)
b=np.array((list7))
print(b)
b=b.reshape(5,1)
print(b)
a=np.hstack((a,b))
print(a)
```

```
[[1 2]
 [2 3]
 [4 5]
 [5 8]
 [6 9]]
[[1 2]
 [2 3]
 [5 8]
 [6 9]]
[[ 1  2]
 [ 2  3]
 [ 5  8]
 [ 6  9]
 [ 8 10]]
[2 3 4 5 6]
[[2]
```

```
[3]
[4]
[5]
[6]]
[[ 1  2  2]
 [ 2  3  3]
 [ 5  8  4]
 [ 6  9  5]
 [ 8 10  6]]
```

In [40]:

```
import numpy as np

# Encoded Data (20 Rows, 5 Columns)
# Columns: [houseid, housetype, price, units, location]
# housetype: Flats=0, Apartments=1, Independent=2
# location: Delhi=0, Gurgaon=1, Noida=2, Faridabad=3, Greater Noida=4, Ghazia

real_estate_data = np.array([
    [1, 0, 45, 10, 0],
    [2, 1, 60, 8, 1],
    [3, 2, 85, 4, 2],
    [4, 0, 50, 12, 3],
    [5, 1, 70, 6, 4],
    [6, 2, 90, 3, 5],
    [7, 0, 55, 9, 0],
    [8, 1, 68, 7, 2],
    [9, 0, 49, 11, 1],
    [10, 2, 88, 5, 4],
    [11, 0, 53, 10, 5],
    [12, 1, 62, 9, 3],
    [13, 0, 47, 13, 0],
    [14, 2, 83, 4, 1],
    [15, 1, 69, 8, 2],
    [16, 0, 52, 12, 3],
    [17, 1, 71, 7, 4],
    [18, 2, 95, 2, 5],
    [19, 0, 54, 11, 2],
    [20, 1, 73, 6, 0]
])
revenue=np.sum(real_estate_data[:,2])
print(revenue)
totalflats=np.sum(real_estate_data[:,3])
print(totalflats)
```

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In [46]:

```
flats = real_estate_data[real_estate_data[:, 1] == 0]
print("Properties in Noida:\n", flats)
flatsrev=np.sum(flats[:,2])
print(flatsrev)

appartment = real_estate_data[real_estate_data[:, 1] == 1]
print("Properties in Noida:\n", appartment)
appartmentrev=np.sum(appartment[:,2])
print(appartmentrev)
```

Properties in Noida:

```
[[ 1  0 45 10  0]
 [ 4  0 50 12  3]
 [ 7  0 55  9  0]
 [ 9  0 49 11  1]
 [11  0 53 10  5]]
```

```
[13  0 47 13  0]
[16  0 52 12  3]
[19  0 54 11  2]]
405
Properties in Noida:
[[ 2  1 60  8  1]
 [ 5  1 70  6  4]
 [ 8  1 68  7  2]
 [12  1 62  9  3]
 [15  1 69  8  2]
 [17  1 71  7  4]
 [20  1 73  6  0]]
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```

In []: